

The project is an opportunity to apply what you have learned in the Deep Learning and Applications course. The project involves visual data in some form as input of the implemented algorithm.

As first you should submit a project proposal (email:puglisi@unica.it) reporting the following information:

- The problem to be investigated.
- The employed data.
- The algorithm exploited to solve the problem.

Such as example:

- **Project 1**

The problem to be investigated: gender recognition from facial images.

The employed data: <https://mmlab.ie.cuhk.edu.hk/projects/CelebA.html>

The algorithm exploited to solve the problem: custom and finetuned CNN (es. MobileNet, Xception).

- **Project 2**

The problem to be investigated: Image-based indoor localization.

The employed data: <https://iplab.dmi.unict.it/MLC2018/>

The algorithm exploited to solve the problem: custom and finetuned CNN (es. MobileNet, Xception).

You can work in teams of up to 3 people. However, results achieved by projects involving 3 people should be more impressive than ones achieved by projects done with fewer people. Note that, although working in team is encouraged, you may also work alone.

Submission

Once you have completed the project, you must submit a final report as a PDF structured as follows:

- Introduction: describe the considered problem and why it is important.
- Data: Describe the data you have employed in the project.
- Methods: Discuss the approach you have implemented to solve the problem described in the introduction.
- Experiments: Discuss the experiments conducted to demonstrate the effectiveness of your approach. As example, you can perform experiments with different hyperparameters, discussion of failure cases, etc. To illustrate the experimental results you should include graphs, tables, figures.
- Conclusion: Summarize the key results and suggest ideas for future extensions.

Note that the project must be also presented during the oral exam.